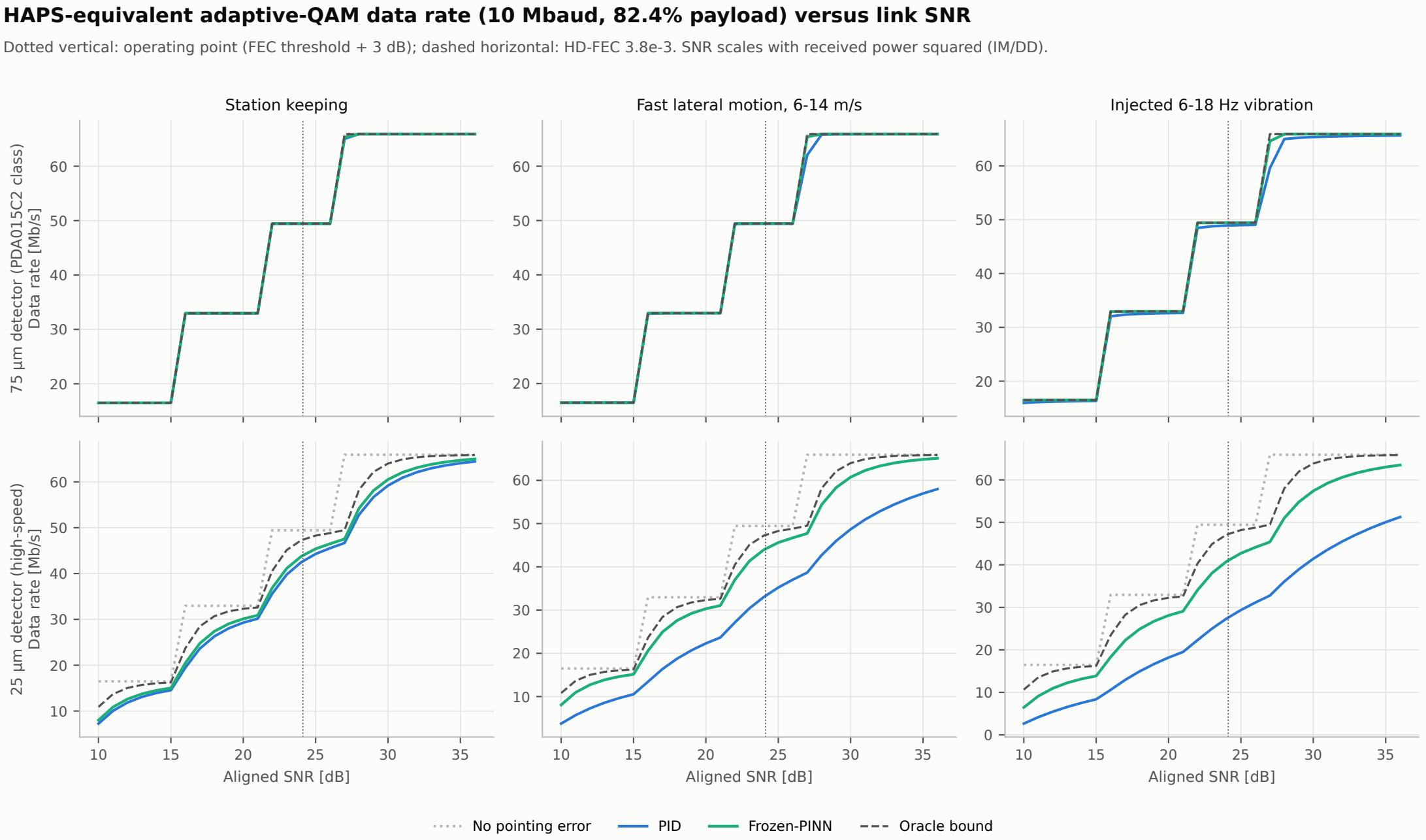
HAPS-equivalent adaptive-QAM data rate (10 Mbaud, 82.4% payload) versus link SNR
Dotted vertical: operating point (FEC threshold + 3 dB); dashed horizontal: HD-FEC 3.8e-3. SNR scales with received power squared (IM/DD).
Station keeping
Fast lateral motion, 6-14 m/s
Injected 6-18 Hz vibration
75 µm detector (PDA015C2 class)
Data rate [Mb/s]
25 µm detector (high-speed)
Data rate [Mb/s]
Aligned SNR [dB]
No pointing error
PID
Frozen-PINN
Oracle bound